

Santhosh Gowru

Data Engineer

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SUMMARY

Data Engineer with 5+ years of experience architecting and optimizing large-scale, end-to-end ETL and machine learning ecosystems. Expert in cloud-native platforms (AWS, Azure), distributed data processing (Spark, Databricks), and ML orchestration frameworks (Airflow, MLflow, Kubeflow). Adept at designing high-throughput data pipelines and deploying production-grade machine learning models that enhance predictive accuracy and operational efficiency. Proven track record of collaborating across data science and engineering teams to deliver intelligent, automated, and business-aligned AI solutions.

SKILLS

Programming & Scripting: Python (NumPy, Pandas, PySpark), SQL, Scala, Java, Shell Scripting, JavaScript, TypeScript
Data Engineering & ETL: ETL/ELT Pipeline Design, Data Ingestion & Transformation, Data Modeling (star/snowflake schema), Data Integration, Batch & Real-Time Processing, Streaming (Kafka, Kinesis), Data Frameworks (Great Expectations, Deequ)
Big Data & Processing Frameworks: Apache Spark, Databricks, Hadoop, Hive, Presto, dbt, Airflow
Databases & Storage: Relational Databases (PostgreSQL, MySQL, SQL Server), NoSQL (MongoDB, DynamoDB), Data Lakes (Delta Lake, Lakehouse), Query Optimization, Indexing, Partitioning
Cloud & Infrastructure: AWS (S3, Glue, Lambda, Redshift, EKS), GCP (BigQuery, Dataflow, Composer), Azure (Data Factory, Synapse), Terraform, CloudFormation, CI/CD (Jenkins, GitHub Actions)
DevOps & Orchestration: Docker, Kubernetes (EKS, GKE, AKS), Helm, ArgoCD, Infrastructure as Code (IaC), Automated Deployment, System Monitoring (Prometheus, Grafana, Datadog)
Machine Learning & MLOps: MLflow, Kubeflow, AWS SageMaker, Vertex AI, Feature Engineering, Feature Stores (Feast, Tecton), Model Deployment, A/B Testing, MLOps Automation
Data Analytics: Snowflake, Looker, Tableau, Excel, Power BI, Data Visualization, KPI Design, Query Optimization
Data Governance & Security: Data Lineage & Cataloging (Collibra, Amundsen), Access Control, Encryption, GDPR Compliance
Methodologies & Soft Skills: Agile/ Scrum, Cross-Functional Collaboration, Scalable Architecture Design, Automation & Optimization, Data-Driven Decision Making, Stakeholder Communication, Continuous Improvement

EXPERIENCE

Data Engineer, Berkshire Hathaway Dec 2024 – Present

- Designed and deployed large-scale ETL pipelines using Apache Spark and Databricks, integrating heterogeneous datasets with 99.8% reliability and achieving zero schema drift across financial, operational, and risk domains.
- Architected a real-time data streaming platform with Kafka and AWS Kinesis, reducing downstream data latency from 15 minutes to under 2 minutes and enabling faster analytics updates.
- Built and optimized Delta Lake storage on AWS S3, enabling versioned data access and partitioned querying that improved analytics query performance by 42% and reduced retrieval latency for business dashboards.
- Operationalized multiple ML models using MLflow and Kubeflow, automating retraining and deployment pipelines that reduced manual release cycles by 70% and improved production uptime to 99.9%.
- Established CI/CD pipelines via GitHub Actions and Terraform, achieving 100% reproducible environments and accelerating infrastructure deployment cycles by 50% while enhancing audit compliance.
- Collaborated with data science and finance teams to build predictive risk models, improving credit risk forecasting accuracy by 12% and enhancing investment decision-making efficiency.
- Monitored and tuned Spark job performance using AWS EMR metrics and Datadog, reducing average job runtime by 30% through dynamic resource allocation and configuration optimization.

Data Engineer, Draxo Infotech Jul 2021 – Jun 2023

- Engineered modular batch and streaming data pipelines using Azure Data Factory and Synapse Analytics, orchestrating ingestion of 2TB+ daily from structured and unstructured sources across departments.
- Developed dbt-based transformation layers integrated with Great Expectations, improving data trustworthiness and eliminating 100% of manual validation previously performed by analysts.
- Migrated on-premise Hadoop workloads to AWS Glue, S3, and Redshift, enhancing scalability and reducing infrastructure costs by 30% via serverless architecture adoption.
- Designed optimized data models (star and snowflake schemas) and applied query tuning to improve analytics response times by 45%, supporting rapid data-driven decision-making.
- Automated job scheduling, monitoring, and alerting using Airflow DAGs, Prometheus, and Grafana, enabling proactive failure resolution and achieving zero missed critical data loads.
- Built metadata-driven ingestion frameworks supporting 25+ data sources, including APIs and databases, implementing dynamic schema detection and complete lineage tracking for transparency.
- Enhanced team productivity by introducing Terraform-based infrastructure provisioning, reducing environment setup time from 4 hours to 15 minutes and supporting 10+ concurrent projects.
- Partnered with BI teams to transform ingestion data into curated Power BI datasets, improving dashboard refresh frequency by 60% and enabling real-time visibility for 200+ stakeholders.

Data Analyst, Bright mind Technologies Jan 2019 – Jun 2021

- Designed and implemented robust data collection, cleaning, and transformation processes using Python (Pandas, NumPy) and SQL, ensuring consistent dataset readiness for analytical and reporting workflows.
- Built interactive Tableau and Power BI dashboards, providing executives with real-time insights into sales, operations, and marketing performance that improved decision turnaround time by 40%.
- Developed automated data pipelines using Python and SQL stored procedures, refreshing reports daily and reducing manual reporting time by 8 hours weekly for analytics teams.
- Created KPI frameworks and validation logic to track performance metrics, ensuring 99.9% data accuracy across multi-department dashboards and quarterly performance reviews.
- Collaborated with cross-functional teams to translate analytical findings into data-driven actions, improving campaign conversion rates by 11% and increasing operational efficiency through process redesign.
- Conducted ad-hoc analyses in SQL and Excel, identifying resource allocation inefficiencies and supporting strategic business decisions that led to 5% annual cost savings.

EDUCATION

Masters of Information Technology Management Sep 2023 – May 2025
Webster University and San Antonio

Bachelors of Computer Science and Engineering Jul 2017 – Sep 2021
Jawaharlal Nehru Technological University, Hyderabad

CERTIFICATION

Microsoft Certified: Power Platform Functional Consultant Associate